

`df1` which had a `NaN` for the column `aqi`.

```
###-STEP 3:- Clean the DataFrame
df['aqi'] = pd.to_numeric(df.aqi,
                          errors='coerce') # Invalid parsing to NaN
print('with NaN->', df.shape) # Comes out as (152, 4)
# Remove NaN (Not a Number) entries in col
df1 = df.dropna(subset = ['aqi'])
print('without NaN->', df1.shape) # (144, 4)
```

## Step 4: Plotting the heat map

We can first use the `Map` class of the `Folium` library to create a map. We can have a map with a particular centre and a particular zoom level. Note that `Folium` creates interactive maps; so even if you want to move to a different map centre or a different zoom level, you can do so by interacting with the map generated. The signature of the `Map` class is given below:

```
class folium.folium.Map(location=None, width='100%',
                        height='100%', left='0%', top='0%', position='relative',
                        tiles='OpenStreetMap', attr=None, min_zoom=0, max_zoom=18,
                        zoom_start=10, min_lat=- 90, max_lat=90, min_lon=-
                        180, max_lon=180, max_bounds=False, crs='EPSG3857',
                        control_scale=False, prefer_canvas=False, no_touch=False,
                        disable_3d=False, png_enabled=False, zoom_control=True,
                        **kwargs)
```

The `Folium` library provides a class `HeatMap`, which can construct a heat map. The signature of this class

is as follows:

```
class folium.plugins.HeatMap(data, name=None, min_
opacity=0.5, max_zoom=18, max_val=1.0, radius=25, blur=15,
gradient=None, overlay=True, control=True, show=True,
**kwargs)
```

The two parameters of importance are `data` and `max_val`. The other parameters are only for changing the look of the heat map. For the `data` parameter, we will provide a `DataFrame` with three columns, namely, `latitude`, `longitude` and `aqi`. So we will create a new `DataFrame` `df2` which has only three columns. For the parameter `max_val`, we will provide the maximum value in the `aqi` column (because the `heat` of the map will depend upon the value of this column). The other parameters are all by default and can be ignored:

```
###-STEP 4: Make folium heat map
init_loc = [23, 77] # Approx over Bhopal
max_aqi = int(df1['aqi'].max())
print('max_aqi->', max_aqi)
m = folium.Map(location = init_loc, zoom_start = 5)

heat_aqi = HeatMap(df2, min_opacity = 0.1, max_val = max_aqi,
                   radius = 20, blur = 20, max_zoom = 2)
m.add_child(heat_aqi)
m # Show the map
```

The heat map on Jupyter notebook looks somewhat like what's shown in Figure 1.

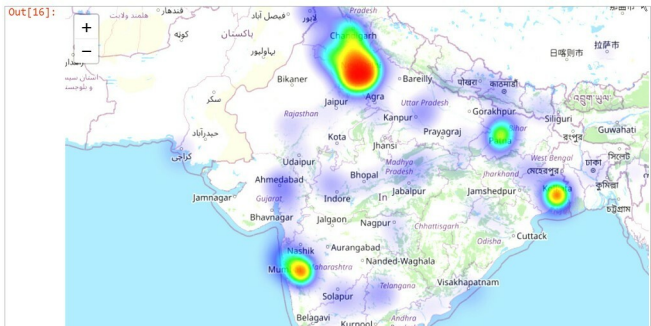


Figure 1: Heat map on Jupyter notebook